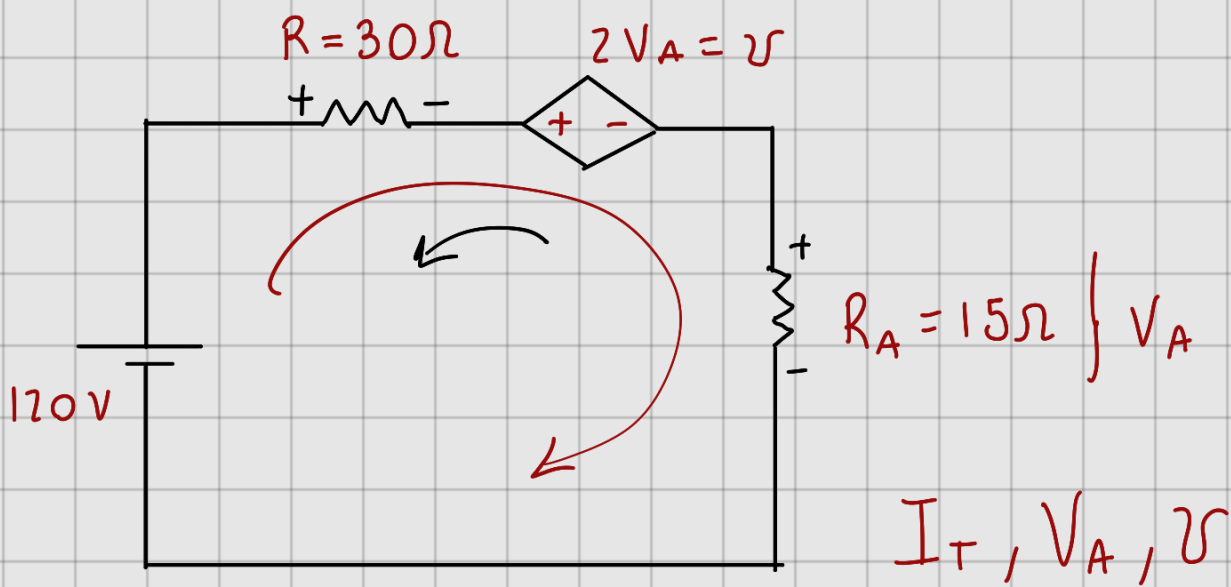




$$120 - V_R - 2V_A - V_A = 0$$

$$120 - RI_T - 2R_A I_T - R_A I_T = 0$$

$$120 - 30I_T - 3(15)I_T = 0$$

$$120 - 30I_T - 45I_T = 0$$

$$-75I_T = -120$$

$$I_T = \frac{-120}{-75} = 1.6A$$

$$V_R = RI_T = 30(1.6A) = 48V$$

$$V_A = R_A I_T = 15(1.6A) = 24V$$

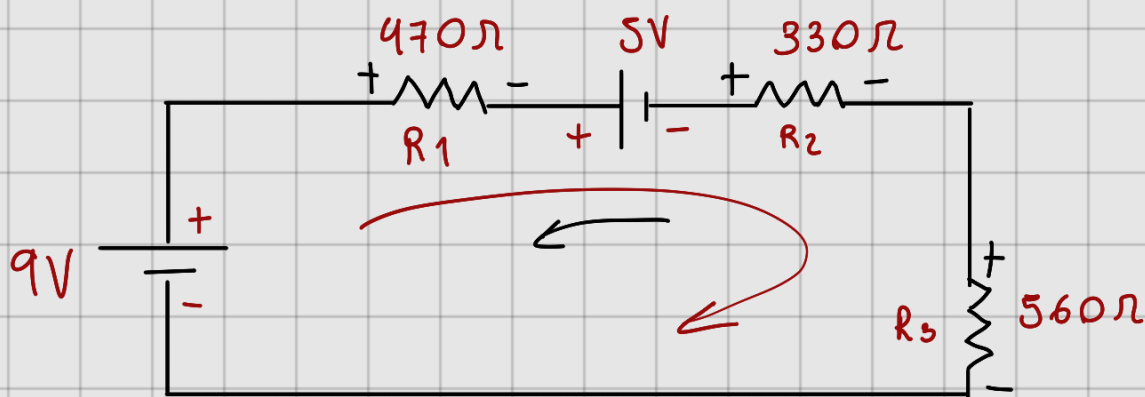
$$2V_A = 2(24V) = 48V$$

$$120 - 48V - 48V - 24V = 0$$

$$I_T = 1.6A$$

$$V_A = 24V$$

$$v = 48V$$



$$9V - V_{R_1} - 5V - V_{R_2} - V_{R_3} = 0$$

$$9V - R_1 I_T - 5V - R_2 I_T - R_3 I_T = 0$$

$$9V - 470 I_T - 5V - 330 I_T - 560 I_T = 0$$

$$4V - 1360 I_T = 0$$

$$-1360 I_T = -4V$$

$$I_T = \frac{-4}{-1360} = 2.9 \text{ mA}$$

$$V_{R_1} = I_T R_1 = 1.363 \text{ V}$$

$$V_{R_2} = I_T R_2 = 0.957 \text{ V}$$

$$V_{R_3} = I_T R_3 = 1.624 \text{ V}$$

$$9V - 1.363V - 5V - 0.957V - 1.624V = 0$$

$$9V - 8.944 \approx 0$$